N.B.K.R.I.S.T ENGINEERING COLLEGE
VIDYANAGAR

AUTONOMUS

SBI CREDIT CARD PROCESSING APPLICATION

SBI CREDIT CARD PROCESSING APPLICATION SYSTEM USING QUEUE IN C

UNDER GUIDENCE

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OBJECTIVE

- ✅ To simulate a real-world **credit card application processing system** using data structures.
- ✅ To implement a **queue-based approach** for handling multiple credit card applications in order.
- ✅ To apply **linked list concepts** for dynamic memory allocation and flexibility in managing applications.
- ✅ To allow users to **apply, view, and process** credit card applications efficiently.
- ✅ To enhance understanding of **queue operations** such as enqueue, dequeue, and peek using C programming.
- ✅ To build a **simple command-line menu interface** for interacting with the system.

WHY C AND DATA STRUCTURE?

- ⚙️ **C is close to hardware** – it allows precise control over memory, which is essential for understanding how data structures work internally.
- 💡 **C is foundational** – it is widely used in academia to teach core programming and data structure concepts clearly and efficiently.
- 🧠 **Data Structures (DSA)** are critical for problem-solving – they teach how to store, access, and manage data logically and efficiently.
- 📚 Mastery of **DSA in C** strengthens your ability to build real-world systems like databases, operating systems, and applications like this credit card system.
- 🔁 **Queue implementation in C** helps understand dynamic memory management using pointers and linked lists.
- 🚀 **C with DSA** provides strong groundwork for competitive programming, system programming, and interviews

📚 Data Structures and Algorithms (DSA)

🔶 **Data Structures**

A way to organize and store data so it can be accessed and modified efficiently.

•Examples:

- •**Array** – fixed-size collection of elements
- •**Linked List** – dynamic list using nodes
- •**Stack & Queue** – LIFO and FIFO structures
- •**Tree & Graph** – hierarchical and network models

✅ **Why DSA Matters**

•Enhances problem-solving skills

•Optimizes code performance

•Essential for technical interviews and real-world systems

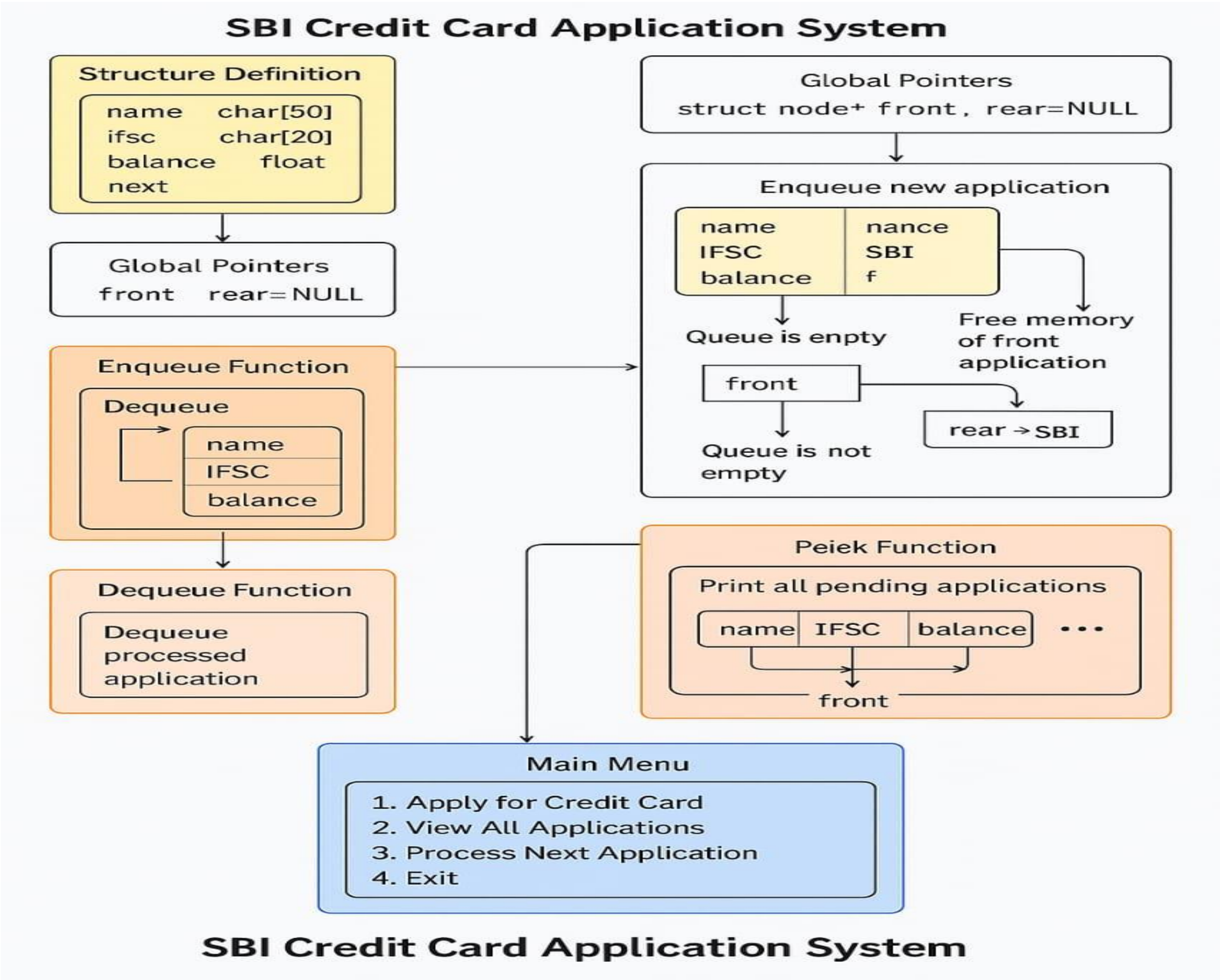
🔷 Algorithms

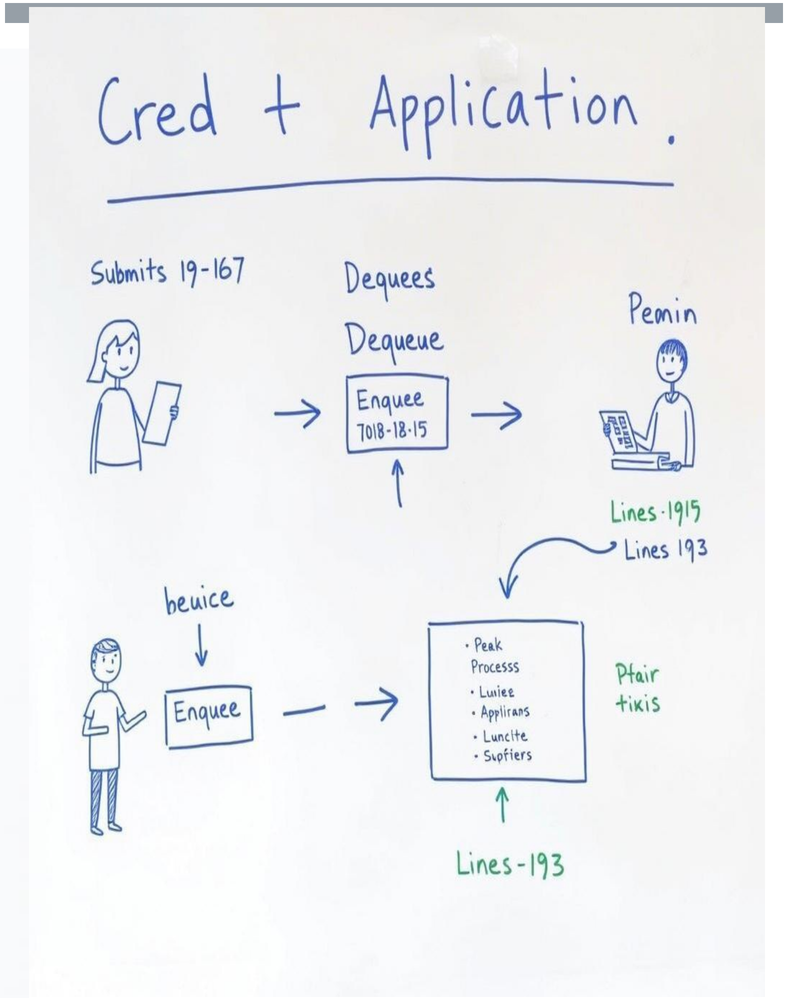
A step-by-step method to solve a problem or perform a task.

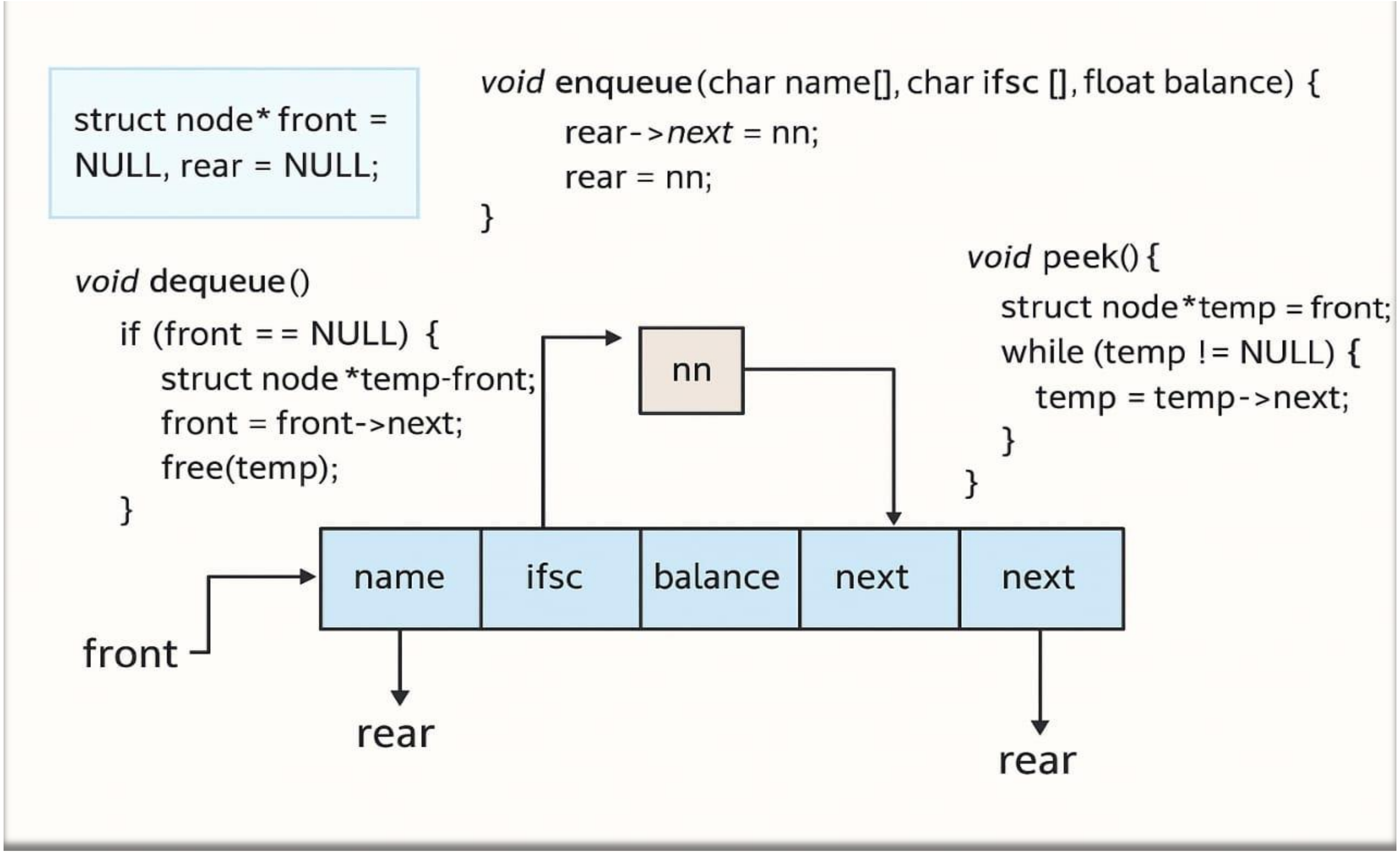
•Characteristics:

- •**Correctness**
- •**Efficiency (Time & Space)**
- •**Clarity and Simplicity**

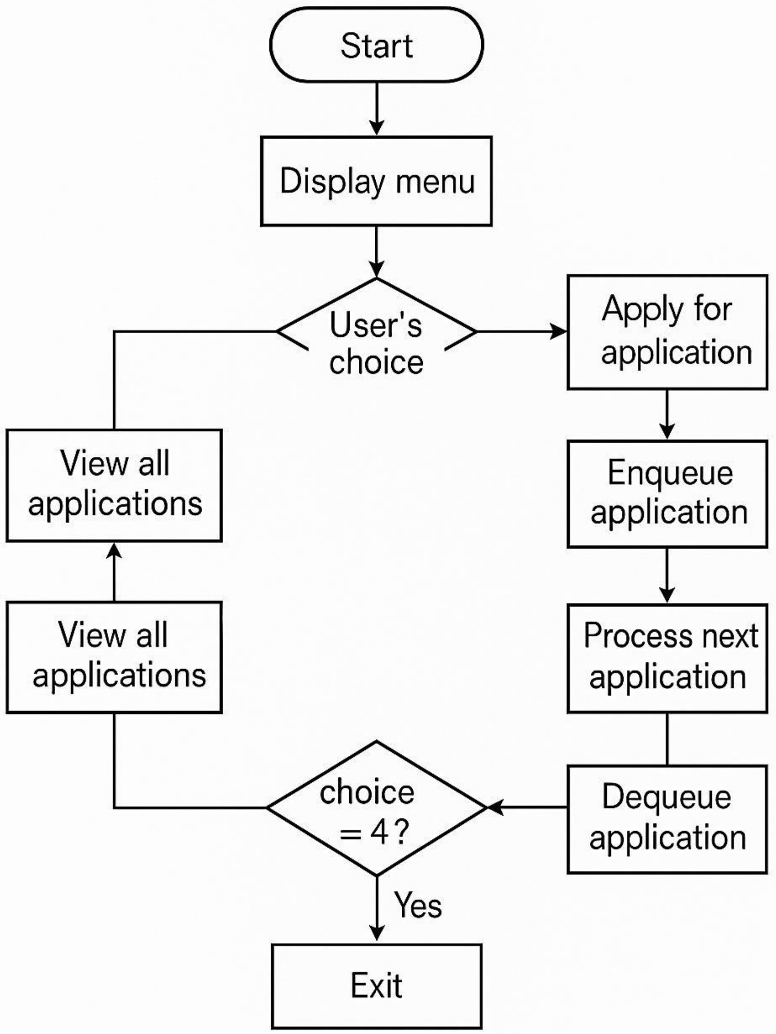
Diagramatic view of the application





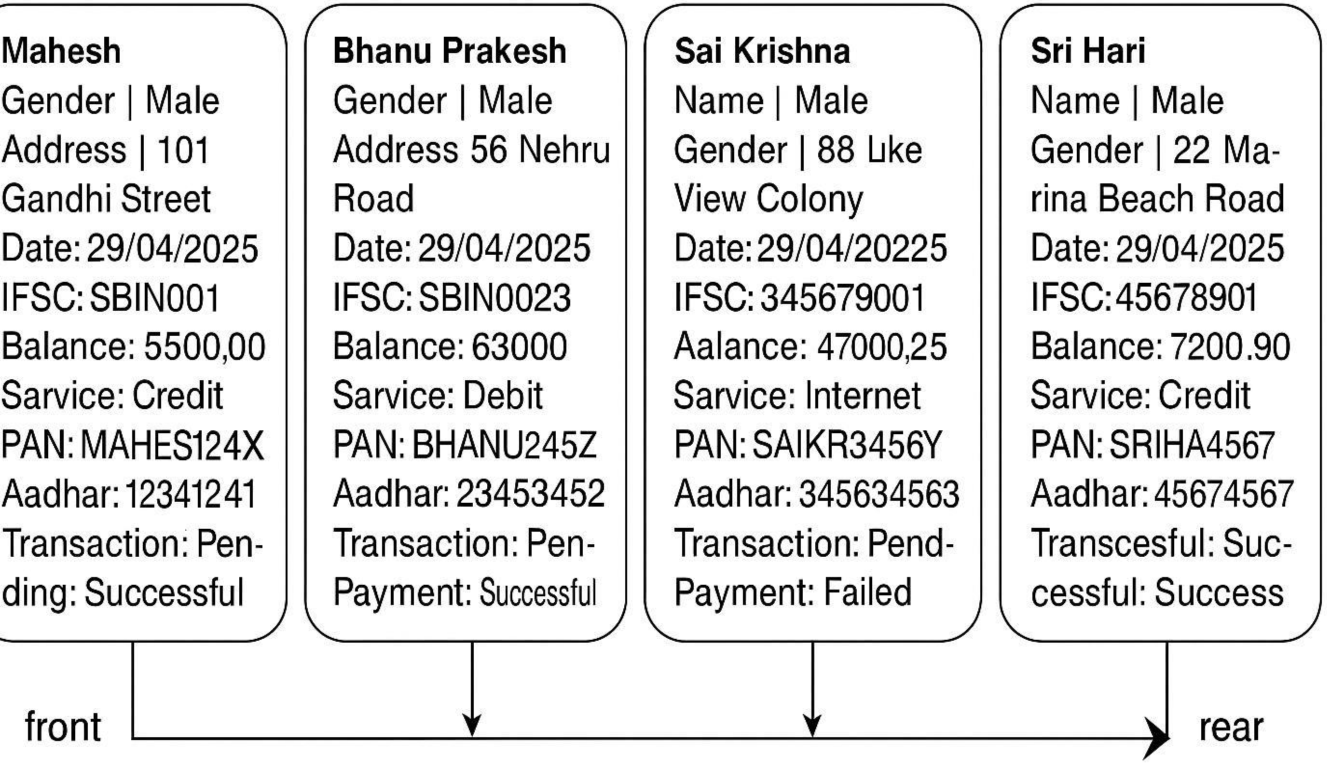

struct node* front =
NULL, rear = NULL;
void enqueue(char name[], char ifsc [], float balance) {
rear->next = nn;
rear = nn;
}
void dequeue()
if (front == NULL) {
struct node *temp-front;
front = front->next;
free(temp);
}
void peek() {
struct node*temp = front;
while (temp != NULL) {
temp = temp->next;
}
}
nn
name
ifsc
balance
next
next
front
rear
rear

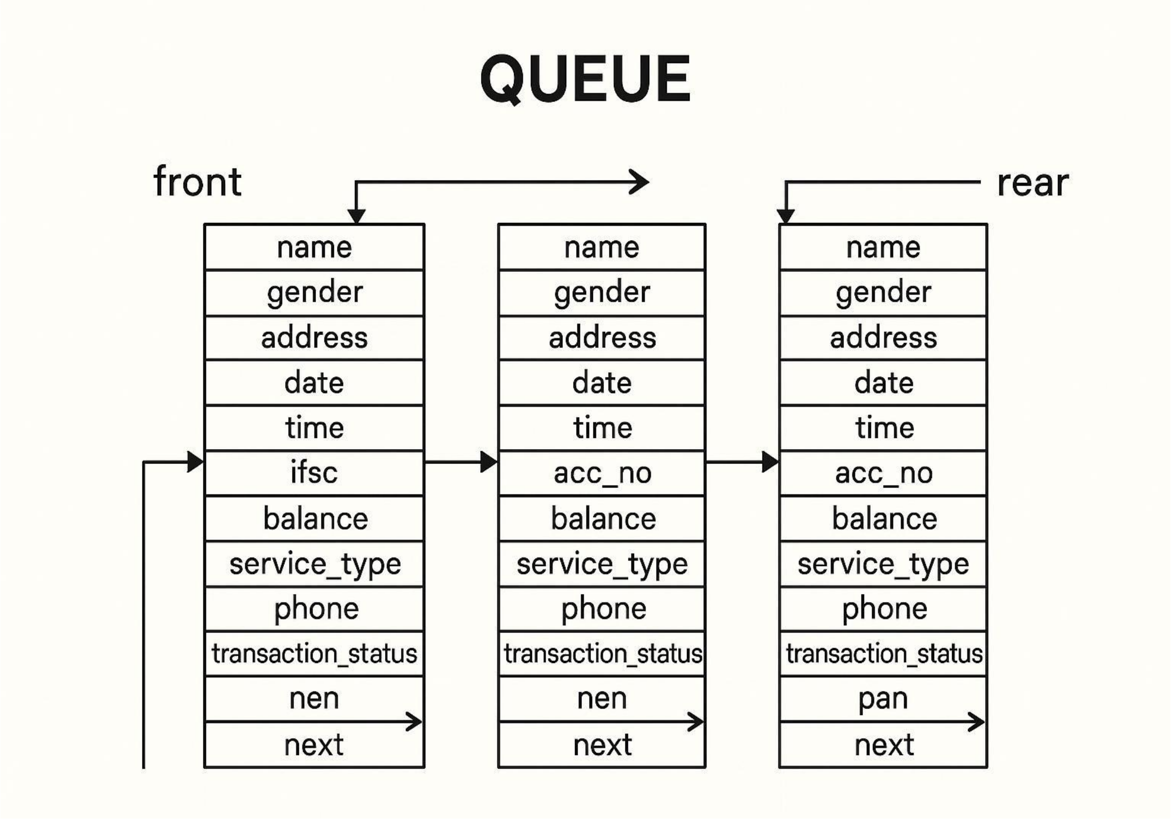
ALGORTHIM AND FLOW CHART



Credit Card Application System

EXAMPLE OUTPUT IN DIAGRAMTIC





THANK YOU